

Case Study #3: HIV Prevention Techniques in Botswana

Tomoko Hida

2024-12-13

Introduction

Between 2013 to 2018, the Botswana Combination Prevention Project (BCPP) was conducted in order to evaluate the efficacy of a community based approach to combating HIV in Botswana. A “combination prevention” package including a combination of individually effective prevention strategies, including testing and treatment services was developed and randomly distributed to 15 of the 30 communities that agreed to participate in the trial, and the other 15 communities proceeded with the “standard of care” standard HIV prevention approach. In order to compare the frequency of new HIV infections between the two prevention strategies, 7,826 HIV negative members of the 30 communities were followed over the course of three years. The following analysis is aimed at answering how the odds of contracting HIV change depending on whether or not study participants live in an intervention community, whether this relationship changes after adjusting for other important predictors of HIV infection, such as gender, marital status, or employment status, and whether the intervention prevention strategy appears to be effective at all. In order to conduct this analysis, we will be evaluating whether the study participant being in the intervention or standard of care community (random_arm) is a statistically significant predictor of the individual’s HIV status after one year in the study (hiv_status_current).

Data Description & Exploratory Data Analysis

The data comes from the first of the three year follow-up period of the study participants in the Botswana Combination Prevention Project. Although three particular variables, community_engagement, alcohol_lastsex, and condom_lastsex, had instances of NA values, none of these variables were used in analyzing the impact of the study participant being in the intervention or standard of care community group.

Table 1: Summary Statistics for Study Participant Intervention Community

Intervention	Standard Care
3875	3951

Table 2: Summary Statistics for Study Participant Gender

F	M
4764	3062

Table 3: Summary Statistics for Study Participant Marital Status

Divorced/separated	Married	Single/never married	Widowed
107	1346	6162	211

Table 4: Summary Statistics for Study Participant Employment Status

Employed	Unemployed looking for work	Unemployed not looking for work
2317	3224	2285

Table 5: Summary Statistics for Study Participant Age at Interview

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
16	22.4	29.7	33.55229	43.1	64.9

Table 6: Summary Statistics for Study Participant HIV Status One Year in Trial

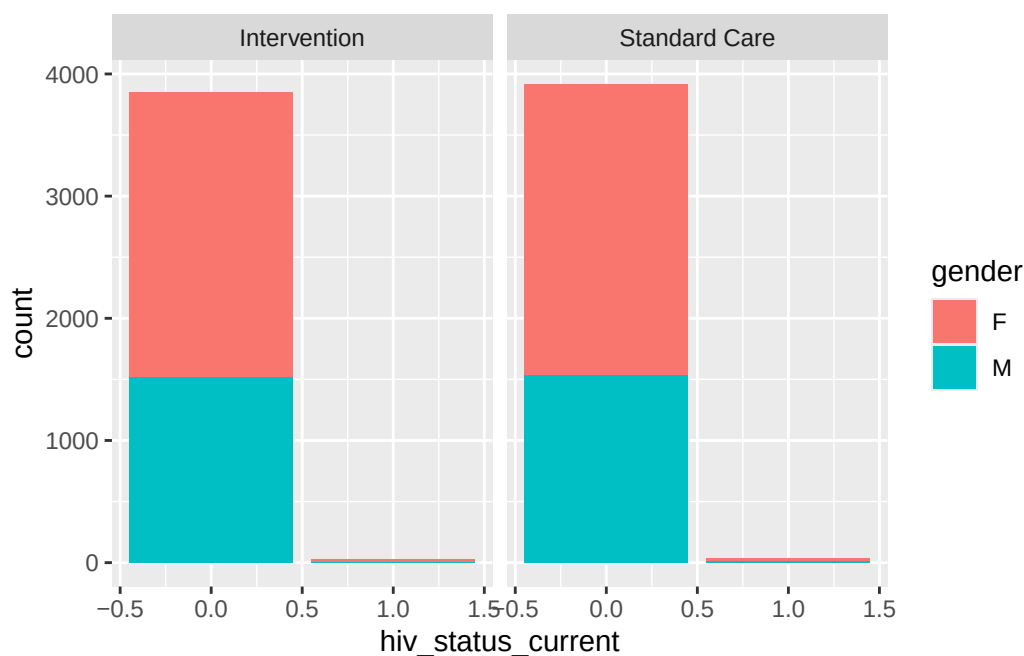
HIV-infected	HIV-uninfected
59	7767

random_arm	# uninfected	# infected
Intervention	3850	25
Standard Care	3917	34

The dataset includes data from 7,826 individuals, and no observations were removed. The distribution of individuals who were in the intervention and standard of care communities were roughly equal, with 3,875 individuals in the study from intervention communities and 3,951 in the standard of care communities. Within all study participants, 4,764 individuals were female, and 3,062 were male, and 107 indicated being divorced/separated, 1,346 indicated being married, 6,162 indicated being single/never married, and 211 indicated being widowed. As for employment status of the study participants, 2,317 indicated being employed, 3,224 indicated being unemployed and looking for work, and 2,285 indicated being unemployed and not looking for work. The minimum study participant age at the time of interview was 16, and the median age was 29.7, and the maximum age was 64.9. As for the total HIV infected individuals and HIV uninfected individuals at the end of the study, 59 were infected and 7767 were uninfected after the trial’s first year, and 25 of the 59

infected individuals were part of the intervention community, and 34 of the 59 infected individuals were part of the standard of care community.

hiv_status_current	# M	# F
0	3050	4717
1	12	47



The histogram above demonstrates the distribution of gender for HIV infected and uninfected individuals, given that the individuals were a part of the intervention community or standard of care community. The column to the left in each faceted histogram is the uninfected group. The proportion of female study participants is greater than that of the male participants for both treatment communities, but for the intervention community, the HIV infected individuals were proportionally more female than male indicated by the red color of the histogram, and for the standard of care community, the HIV infected individuals look to be identifiably both male and female, indicated by the blue and red.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.008605	0.001376	6.2530	0.0000
random_arm	-0.002154	0.001956	-1.1013	0.2708

Statistical Methods

The modeling process for finding an appropriate combination of the relevant explanatory variables for our response variable of interest `hiv_current_status` was conducted using a forward selection variable selection technique, using AIC as the metric for comparison. The model that our analysis uses is such that the explanatory variable of interest, `random_arm`, must have been included, so

our model includes the variables for whether the participant was in an intervention or standard of care community (*random_arm*), whether the participant was Male or Female (*gender*), whether the participant was divorced, married, single, or widowed (*marital_status*), as well as whether the participant was employed, unemployed and looking, or unemployed and not looking (*employment_status*). Below is the population model for our logistic regression.

$$\begin{aligned} \text{logit}(\pi(HIVstatus_i)) = & \beta_0 + \beta_1(randomArm_i) + \beta_2(gender) + \beta_3(married_i) + \beta_4(single_i) \\ & + \beta_5(widowed_i) + \beta_6(unemployedLooking_i) + \beta_7(unemployedNotLooking_i) \end{aligned}$$

Let $HIVstatus_i$ denote the HIV infection status of the i – th study participant after one year in the study (categorized as HIV-infected (1), or HIV-uninfected (0)), $randomArm_i$ denote whether or not they were a part of a intervention community or not (intervention (1), standard of care (0)), $gender_i$ denote their gender (categorized as either male (1) or female (0)), $married_i$ denote whether they are married (married(1) or not married(0)), $single_i$ denote whether they are single (single(1) or not single(0)), $widowed_i$ denote whether they are widowed (widowed(1) or not widowed(0)), $unemployedLooking_i$ denote whether they are unemployed and looking for employment (unemployed and looking (1) or unemployed and not looking(0)), and $unemployedNotLooking_i$ denote whether they are unemployed and not looking(unemployed and not looking (1) or not unemployed and not looking(0)).

The model that considers only the intervention treatment variable to estimate the current HIV status of a study participant produces the following parameter estimate, seen in the table below. This indicates the exponentiating value for the odds of contracting HIV given that a study participant is in an intervention treatment community.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.008605	0.001376	6.2530	0.0000
random_arm	-0.002154	0.001956	-1.1013	0.2708

The table below indicates the estimates of our logistic regression parameters, and their accompanying standard errors as well as their respective statistical significance as measured by the p-values.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.020029	0.008556	2.3409	0.0193
random_arm	-0.001726	0.001959	-0.8813	0.3782
gender	-0.006302	0.002028	-3.1070	0.0019
marital_statusMarried	-0.016207	0.008679	-1.8673	0.0619
marital_statusSingle/never married	-0.009933	0.008438	-1.1772	0.2392
marital_statusWidowed	-0.020150	0.010270	-1.9620	0.0498
employment_statusUnemployed looking for work	0.004380	0.002391	1.8319	0.0670
employment_statusUnemployed not looking for work	0.000610	0.002565	0.2378	0.8121

Results and Discussion

The study objective was to understand the efficacy of the Botswana Combination Prevention Project (BCPP), one year into the trial. Among the 30 communities across Botswana that participated in the Botswana Combination Prevention Project (BCPP), the communities that received the “combination prevention” package as opposed to the communities that proceeded with their standard HIV prevention approach “standard of care” were associated with an estimated 0.9978 times change (0.22% decrease) in the odds of contracting HIV. After adjusting for other important predictors of HIV infection, determined to be the variables gender, marital status, and employment status, the odds of contracting HIV depending on whether or not study participants lived in an intervention community stayed generally the same. With the model that accounts for study participants’ gender, marital status, and employment status, living in an intervention community was associated with an estimated 0.9983 times change (0.17% decrease) in the odds of contracting HIV. Considering that the p-value of the `random_arm` variable coefficient is 0.3782, we fail to reject the null hypothesis that the intervention has no effect at all in reducing the probability of new HIV infection within the first year after its implementation.

Although the data collected for this analysis did have some missing values across three variables, the set of variables used for this study were robust in their balanced frequency of observations per level of factor, and for the sheer number of different variables observed. The limitations of this study, however, may be the types of variables observed—many of the variables seem to explain similar portions of the response data, such as smoking habits, alcohol habits, and alcohol and contraceptive habits during sex. The data used for this analysis, however, does put forth the clear conclusion that the BCPP seems to be not statistically significant in reducing HIV contraction after one year of the trial.

R Appendix

```
#loading packages
library(tidyverse)
library(dplyr)
library(broom)
library(kableExtra)

# Loading the case study data
bcpp <- read_csv("bcpp-year-one.csv")

# checking for NA values
colSums(is.na(bcpp))
```

de_subj_idC	random_arm	hiv_status_current
0	0	0
gender	age_at_interview	community_engagement
0	0	275
religious_affil	marital_status	education
0	0	0
employment_status	partners_lifetime	condom_lastsex
0	0	977
alcohol_lastsex	dx_tb_ever	dx_heartdisease_ever
971	0	0
alcohol	smoke	
0	0	

```
#exploring the data

#numerical ordinal variables
table(bcpp$partners_lifetime) # num ordinal 0 - 500
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
952	1222	1154	1176	734	757	379	220	163	69	366	46	54	12	18	118
16	17	18	19	20	21	22	23	24	25	26	27	28	30	31	32
13	10	8	6	133	6	2	1	1	32	2	3	2	42	1	2
33	34	35	37	39	40	41	44	45	46	50	51	54	55	60	70
1	1	14	1	1	17	2	1	3	1	32	1	1	3	4	6
75	80	86	90	100	101	110	150	200	300	500					
1	1	1	1	21	1	1	1	3	1	1					

```
table(bcpp$age_at_interview) # numeric ordinal 16 to 64.9
```

16	16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	17	17.1	17.2	17.3	17.4	17.5
5	18	21	27	29	19	24	37	26	36	33	27	26	24	34	24
17.6	17.7	17.8	17.9	18	18.1	18.2	18.3	18.4	18.5	18.6	18.7	18.8	18.9	19	19.1
30	31	24	26	24	30	32	32	32	31	33	33	31	22	27	27
19.2	19.3	19.4	19.5	19.6	19.7	19.8	19.9	20	20.1	20.2	20.3	20.4	20.5	20.6	20.7
29	36	33	29	37	35	31	33	48	35	34	25	37	35	28	44
20.8	20.9	21	21.1	21.2	21.3	21.4	21.5	21.6	21.7	21.8	21.9	22	22.1	22.2	22.3
40	34	32	36	20	29	36	36	31	41	22	30	32	27	34	25
22.4	22.5	22.6	22.7	22.8	22.9	23	23.1	23.2	23.3	23.4	23.5	23.6	23.7	23.8	23.9
29	35	27	34	40	37	24	33	38	31	36	34	27	20	25	27
24	24.1	24.2	24.3	24.4	24.5	24.6	24.7	24.8	24.9	25	25.1	25.2	25.3	25.4	25.5
27	33	35	28	26	27	29	23	27	30	38	20	22	30	38	26
25.6	25.7	25.8	25.9	26	26.1	26.2	26.3	26.4	26.5	26.6	26.7	26.8	26.9	27	27.1
39	32	27	20	13	32	23	25	22	20	32	20	25	30	25	28
27.2	27.3	27.4	27.5	27.6	27.7	27.8	27.9	28	28.1	28.2	28.3	28.4	28.5	28.6	28.7
22	29	31	16	18	22	21	29	22	29	24	28	26	20	27	31
28.8	28.9	29	29.1	29.2	29.3	29.4	29.5	29.6	29.7	29.8	29.9	30	30.1	30.2	30.3
21	27	27	28	16	34	14	25	21	21	26	24	18	24	12	24
30.4	30.5	30.6	30.7	30.8	30.9	31	31.1	31.2	31.3	31.4	31.5	31.6	31.7	31.8	31.9
21	19	27	28	19	21	23	23	27	20	25	19	19	18	27	18
32	32.1	32.2	32.3	32.4	32.5	32.6	32.7	32.8	32.9	33	33.1	33.2	33.3	33.4	33.5
31	17	24	22	28	26	13	33	18	22	20	14	24	22	20	16
33.6	33.7	33.8	33.9	34	34.1	34.2	34.3	34.4	34.5	34.6	34.7	34.8	34.9	35	35.1
14	13	18	25	11	23	14	15	22	17	13	21	11	20	18	14
35.2	35.3	35.4	35.5	35.6	35.7	35.8	35.9	36	36.1	36.2	36.3	36.4	36.5	36.6	36.7
12	15	14	13	23	5	13	9	14	16	18	16	15	10	11	6
36.8	36.9	37	37.1	37.2	37.3	37.4	37.5	37.6	37.7	37.8	37.9	38	38.1	38.2	38.3
10	10	14	9	6	9	8	11	14	10	22	7	4	14	17	15
38.4	38.5	38.6	38.7	38.8	38.9	39	39.1	39.2	39.3	39.4	39.5	39.6	39.7	39.8	39.9
9	7	12	7	6	9	14	14	13	11	17	6	11	10	11	7
40	40.1	40.2	40.3	40.4	40.5	40.6	40.7	40.8	40.9	41	41.1	41.2	41.3	41.4	41.5
14	11	10	14	12	11	8	13	10	5	4	13	12	11	4	8
41.6	41.7	41.8	41.9	42	42.1	42.2	42.3	42.4	42.5	42.6	42.7	42.8	42.9	43	43.1
11	10	7	5	7	7	6	10	6	5	10	15	7	6	9	12
43.2	43.3	43.4	43.5	43.6	43.7	43.8	43.9	44	44.1	44.2	44.3	44.4	44.5	44.6	44.7
11	7	6	5	9	5	7	18	10	4	8	12	7	5	4	8
44.8	44.9	45	45.1	45.2	45.3	45.4	45.5	45.6	45.7	45.8	45.9	46	46.1	46.2	46.3
8	4	11	16	7	5	8	9	7	3	12	11	9	7	16	7
46.4	46.5	46.6	46.7	46.8	46.9	47	47.1	47.2	47.3	47.4	47.5	47.6	47.7	47.8	47.9
11	3	8	8	6	6	7	12	9	12	11	7	8	3	5	7
48	48.1	48.2	48.3	48.4	48.5	48.6	48.7	48.8	48.9	49	49.1	49.2	49.3	49.4	49.5
10	11	10	11	10	3	5	14	7	9	8	11	11	11	11	8
49.6	49.7	49.8	49.9	50	50.1	50.2	50.3	50.4	50.5	50.6	50.7	50.8	50.9	51	51.1
14	8	9	5	7	8	8	7	13	11	12	15	8	7	10	8
51.2	51.3	51.4	51.5	51.6	51.7	51.8	51.9	52	52.1	52.2	52.3	52.4	52.5	52.6	52.7
25	5	6	11	10	8	5	11	14	10	11	12	11	4	7	10
52.8	52.9	53	53.1	53.2	53.3	53.4	53.5	53.6	53.7	53.8	53.9	54	54.1	54.2	54.3

9	16	13	16	12	9	6	7	4	9	11	13	11	5	12	10
54.4	54.5	54.6	54.7	54.8	54.9	55	55.1	55.2	55.3	55.4	55.5	55.6	55.7	55.8	55.9
11	5	14	10	7	3	14	16	10	11	8	2	12	11	3	11
56	56.1	56.2	56.3	56.4	56.5	56.6	56.7	56.8	56.9	57	57.1	57.2	57.3	57.4	57.5
19	15	11	8	9	7	8	11	13	7	13	13	17	10	12	4
57.6	57.7	57.8	57.9	58	58.1	58.2	58.3	58.4	58.5	58.6	58.7	58.8	58.9	59	59.1
6	8	9	7	11	12	10	6	12	8	10	7	6	13	12	13
59.2	59.3	59.4	59.5	59.6	59.7	59.8	59.9	60	60.1	60.2	60.3	60.4	60.5	60.6	60.7
16	12	9	7	7	6	5	8	6	11	10	10	5	10	6	7
60.8	60.9	61	61.1	61.2	61.3	61.4	61.5	61.6	61.7	61.8	61.9	62	62.1	62.2	62.3
8	5	7	8	14	7	13	5	15	9	8	9	5	11	18	5
62.4	62.5	62.6	62.7	62.8	62.9	63	63.1	63.2	63.3	63.4	63.5	63.6	63.7	63.8	63.9
9	5	11	3	10	7	11	9	9	6	7	5	9	1	6	6
64	64.1	64.2	64.3	64.4	64.5	64.6	64.7	64.8	64.9						
9	7	6	8	9	3	4	10	5	7						

```
#character binary variables (can be labeled 1 and 0)
table(bcpp$random_arm) # char Intervention / Standard of Care
```

Intervention	Standard of Care
3875	3951

```
table(bcpp$hiv_status_current) # char HIV-infected / HIV-uninfected
```

HIV-infected	HIV-uninfected
59	7767

```
table(bcpp$gender) # char F / M
```

F	M
4764	3062

```
table(bcpp$religious_affil) # char 2 levels
```

Affiliated with religion	No religious affiliation
5328	2498


```
table(bcpp$condom_lastsex) # char Yes / No
```

```

No  Yes
2397 4452

```

```
table(bcpp$dx_tb_ever) # char 2 levels No or unknown / Yes
```

```

No or unknown      Yes
      7716          110

```

```
table(bcpp$dx_heartdisease_ever) # char 2 levels No or unknown / Yes
```

```

No or unknown      Yes
      7740          86

```

```
table(bcpp$smoke) # char 2 levels
```

```

No  Yes
6786 1040

```

```
#character multiple level variables
```

```
table(bcpp$alcohol) # char 5 levels
```

```

      2 to 3 times a week      Less then once a week more than 3 times a week
              550                      523                      294
      Never                      Once a week
      5887                      572

```

```
table(bcpp$alcohol_lastsex) # char 3 levels
```

```

Both of us      My partner      Myself Neither of us
      185          278          220          6172

```

```
table(bcpp$marital_status) # char 4 levels
```

Divorced/separated	Married	Single/never married
107	1346	6162
Widowed		
211		

```
table(bcpp$education) # char 5 levels
```

Higher than senior secondary	Junior secondary
1333	2903
Non-formal	Primary
729	1283
Senior secondary	
1578	

```
table(bcpp$employment_status) # char 3 levels
```

Employed	Unemployed looking for work
2317	3224
Unemployed not looking for work	
2285	

```
table(bcpp$community_engagement) # char 3 levels
```

Not active at all	Somewhat active	Very active
3924	1961	1666

```
#changing from character to binary numeric
bcpp$hiv_status_current <- ifelse(bcpp$hiv_status_current == "HIV-infected", 1,
  ↪ 0)
bcpp$random_arm <- ifelse(bcpp$random_arm == "Intervention", 1, 0)
bcpp$gender <- ifelse(bcpp$gender == "M", 1, 0)
bcpp$religious_affil <- ifelse(bcpp$religious_affil == "Affiliated with
  ↪ religion", 1, 0)
bcpp$condom_lastsex <- ifelse(bcpp$condom_lastsex == "Yes", 1, 0)
bcpp$dx_tb_ever <- ifelse(bcpp$dx_tb_ever == "Yes", 1, 0)
bcpp$dx_heartdisease_ever <- ifelse(bcpp$dx_heartdisease_ever == "Yes", 1, 0)
```

```

bcpp$smoke <- ifelse(bcpp$smoke == "Yes", 1, 0)

#variable selection
intercept_glm <- glm(hiv_status_current ~ random_arm, data = bcpp)
full_model <- glm(hiv_status_current ~ random_arm + age_at_interview + alcohol +
  ↪ dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender +
  ↪ marital_status + partners_lifetime + religious_affil + smoke, data = bcpp)

# First step of the forward selection process
add1(intercept_glm, hiv_status_current ~ random_arm + age_at_interview + alcohol
  ↪ + dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender
  ↪ + marital_status + partners_lifetime + religious_affil + smoke, test = "F")

```

Single term additions

Model:

```

hiv_status_current ~ random_arm

```

	Df	Deviance	AIC	F value	Pr(>F)
<none>		58.546	-16096		
age_at_interview	1	58.530	-16096	2.1772	0.140113
alcohol	4	58.522	-16091	0.7950	0.528238
dx_heartdisease_ever	1	58.541	-16095	0.6370	0.424826
dx_tb_ever	1	58.540	-16095	0.8021	0.370500
education	4	58.481	-16097	2.1624	0.070588 .
employment_status	2	58.497	-16099	3.2764	0.037814 *
gender	1	58.480	-16103	8.8111	0.003003 **
marital_status	3	58.475	-16100	3.1817	0.022912 *
partners_lifetime	1	58.545	-16094	0.2120	0.645232
religious_affil	1	58.545	-16094	0.0849	0.770823
smoke	1	58.542	-16095	0.5043	0.477651

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

step_one <- update(intercept_glm, . ~ . + gender)

add1(step_one, hiv_status_current ~ random_arm + age_at_interview + alcohol +
  ↪ dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender +
  ↪ marital_status + partners_lifetime + religious_affil + smoke, test = "F")

```

Single term additions

Model:

```

hiv_status_current ~ random_arm + gender

```

	Df	Deviance	AIC	F value	Pr(>F)
<none>		58.480	-16103		

age_at_interview	1	58.455	-16104	3.3155	0.06867	.	
alcohol	4	58.437	-16101	1.4341	0.21981		
dx_heartdisease_ever	1	58.474	-16102	0.7980	0.37172		
dx_tb_ever	1	58.475	-16102	0.7569	0.38434		
education	4	58.413	-16104	2.2570	0.06050	.	
employment_status	2	58.437	-16105	2.8920	0.05553	.	
marital_status	3	58.396	-16108	3.7770	0.01010	*	
partners_lifetime	1	58.480	-16101	0.0001	0.99381		
religious_affil	1	58.470	-16102	1.4247	0.23267		
smoke	1	58.479	-16101	0.1147	0.73488		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
step_two <- update(step_one, . ~ . + marital_status)

add1(step_two, hiv_status_current ~ random_arm + age_at_interview + alcohol +
  ↪ dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender +
  ↪ marital_status + partners_lifetime + religious_affil + smoke, test = "F")
```

Single term additions

Model:

		Df	Deviance	AIC	F value	Pr(>F)
<none>			58.396	-16108		
age_at_interview	1	58.395	-16106	0.1041	0.7470	
alcohol	4	58.359	-16105	1.2209	0.2996	
dx_heartdisease_ever	1	58.392	-16107	0.5051	0.4773	
dx_tb_ever	1	58.391	-16107	0.6173	0.4321	
education	4	58.348	-16107	1.5924	0.1733	
employment_status	2	58.365	-16108	2.0774	0.1253	
partners_lifetime	1	58.396	-16106	0.0003	0.9861	
religious_affil	1	58.391	-16107	0.6364	0.4250	
smoke	1	58.396	-16106	0.0146	0.9037	

```
step_three <- update(step_two, . ~ . + employment_status)

add1(step_three, hiv_status_current ~ random_arm + age_at_interview + alcohol +
  ↪ dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender +
  ↪ marital_status + partners_lifetime + religious_affil + smoke, test = "F")
```

Single term additions

Model:

hiv_status_current ~ random_arm + gender + marital_status + employment_status

	Df	Deviance	AIC	F value	Pr(>F)
<none>		58.365	-16108		
age_at_interview	1	58.364	-16106	0.0926	0.7610
alcohol	4	58.332	-16105	1.0979	0.3557
dx_heartdisease_ever	1	58.361	-16107	0.4438	0.5053
dx_tb_ever	1	58.360	-16107	0.5906	0.4422
education	4	58.327	-16106	1.2646	0.2815
partners_lifetime	1	58.365	-16106	0.0060	0.9382
religious_affil	1	58.362	-16107	0.3766	0.5395
smoke	1	58.365	-16106	0.0105	0.9185

```
# model according to stepwise selection with random_arm included guaranteed
↪ (add1()): glm(hiv_status_current ~ random_arm + gender + marital_status, data
↪ = bcpp)

# Forward selection using AIC as the metric
step(intercept_glm, scope = list(lower = intercept_glm, upper = full_model),
↪ direction="forward")
```

Start: AIC=-16096.11

hiv_status_current ~ random_arm

	Df	Deviance	AIC
+ gender	1	58.480	-16103
+ marital_status	3	58.475	-16100
+ employment_status	2	58.497	-16099
+ education	4	58.481	-16097
+ age_at_interview	1	58.530	-16096
<none>		58.546	-16096
+ dx_tb_ever	1	58.540	-16095
+ dx_heartdisease_ever	1	58.541	-16095
+ smoke	1	58.542	-16095
+ partners_lifetime	1	58.545	-16094
+ religious_affil	1	58.545	-16094
+ alcohol	4	58.522	-16091

Step: AIC=-16102.92

hiv_status_current ~ random_arm + gender

	Df	Deviance	AIC
+ marital_status	3	58.396	-16108
+ employment_status	2	58.437	-16105
+ age_at_interview	1	58.455	-16104
+ education	4	58.413	-16104
<none>		58.480	-16103
+ religious_affil	1	58.470	-16102
+ dx_heartdisease_ever	1	58.474	-16102

+ dx_tb_ever	1	58.475	-16102
+ smoke	1	58.479	-16101
+ partners_lifetime	1	58.480	-16101
+ alcohol	4	58.437	-16101

Step: AIC=-16108.25

hiv_status_current ~ random_arm + gender + marital_status

	Df	Deviance	AIC
+ employment_status	2	58.365	-16108
<none>		58.396	-16108
+ religious_affil	1	58.391	-16107
+ dx_tb_ever	1	58.391	-16107
+ dx_heartdisease_ever	1	58.392	-16107
+ education	4	58.348	-16107
+ age_at_interview	1	58.395	-16106
+ smoke	1	58.396	-16106
+ partners_lifetime	1	58.396	-16106
+ alcohol	4	58.359	-16105

Step: AIC=-16108.41

hiv_status_current ~ random_arm + gender + marital_status + employment_status

	Df	Deviance	AIC
<none>		58.365	-16108
+ dx_tb_ever	1	58.360	-16107
+ dx_heartdisease_ever	1	58.361	-16107
+ religious_affil	1	58.362	-16107
+ age_at_interview	1	58.364	-16106
+ smoke	1	58.365	-16106
+ partners_lifetime	1	58.365	-16106
+ education	4	58.327	-16106
+ alcohol	4	58.332	-16105

Call: glm(formula = hiv_status_current ~ random_arm + gender + marital_status +
employment_status, data = bcpp)

Coefficients:

```

(Intercept)
  0.0200285
random_arm
-0.0017262
gender
-0.0063023
marital_statusMarried
-0.0162065

```

```

marital_statusSingle/never married
-0.0099330
marital_statusWidowed
-0.0201497
employment_statusUnemployed looking for work
0.0043805
employment_statusUnemployed not looking for work
0.0006099

```

```

Degrees of Freedom: 7825 Total (i.e. Null); 7818 Residual
Null Deviance: 58.56
Residual Deviance: 58.36 AIC: -16110

```

```

# automated forward selection chooses glm(hiv_status_current ~ random_arm +
  ↪ gender + marital_status + employment_status, data = bcpp)

```

```

# Backward selection using AIC as the metric
step(full_model, direction = "backward")

```

Start: AIC=-16091.68

```

hiv_status_current ~ random_arm + age_at_interview + alcohol +
  dx_heartdisease_ever + dx_tb_ever + education + employment_status +
  gender + marital_status + partners_lifetime + religious_affil +
  smoke

```

	Df	Deviance	AIC
- alcohol	4	58.317	-16095
- education	4	58.318	-16095
- age_at_interview	1	58.281	-16094
- partners_lifetime	1	58.281	-16094
- religious_affil	1	58.281	-16094
- dx_heartdisease_ever	1	58.283	-16093
- employment_status	2	58.299	-16093
- smoke	1	58.285	-16093
- dx_tb_ever	1	58.285	-16093
- random_arm	1	58.285	-16093
<none>		58.281	-16092
- marital_status	3	58.328	-16091
- gender	1	58.353	-16084

Step: AIC=-16094.79

```

hiv_status_current ~ random_arm + age_at_interview + dx_heartdisease_ever +
  dx_tb_ever + education + employment_status + gender + marital_status +
  partners_lifetime + religious_affil + smoke

```

	Df	Deviance	AIC
- education	4	58.354	-16098

- age_at_interview	1	58.317	-16097
- smoke	1	58.317	-16097
- partners_lifetime	1	58.317	-16097
- religious_affil	1	58.319	-16096
- dx_heartdisease_ever	1	58.320	-16096
- dx_tb_ever	1	58.321	-16096
- random_arm	1	58.321	-16096
- employment_status	2	58.337	-16096
<none>		58.317	-16095
- marital_status	3	58.367	-16094
- gender	1	58.381	-16088

Step: AIC=-16097.87

hiv_status_current ~ random_arm + age_at_interview + dx_heartdisease_ever +
 dx_tb_ever + employment_status + gender + marital_status +
 partners_lifetime + religious_affil + smoke

	Df	Deviance	AIC
- partners_lifetime	1	58.354	-16100
- age_at_interview	1	58.354	-16100
- smoke	1	58.354	-16100
- religious_affil	1	58.357	-16100
- dx_heartdisease_ever	1	58.357	-16100
- dx_tb_ever	1	58.358	-16099
- random_arm	1	58.359	-16099
- employment_status	2	58.382	-16098
<none>		58.354	-16098
- marital_status	3	58.406	-16097
- gender	1	58.419	-16091

Step: AIC=-16099.85

hiv_status_current ~ random_arm + age_at_interview + dx_heartdisease_ever +
 dx_tb_ever + employment_status + gender + marital_status +
 religious_affil + smoke

	Df	Deviance	AIC
- age_at_interview	1	58.354	-16102
- smoke	1	58.354	-16102
- religious_affil	1	58.357	-16102
- dx_heartdisease_ever	1	58.357	-16102
- dx_tb_ever	1	58.358	-16101
- random_arm	1	58.359	-16101
- employment_status	2	58.382	-16100
<none>		58.354	-16100
- marital_status	3	58.406	-16099
- gender	1	58.420	-16093

Step: AIC=-16101.84


```
hiv_status_current ~ random_arm + dx_heartdisease_ever + dx_tb_ever +
  employment_status + gender + marital_status + religious_affil +
  smoke
```

	Df	Deviance	AIC
- smoke	1	58.354	-16104
- religious_affil	1	58.357	-16103
- dx_heartdisease_ever	1	58.357	-16103
- dx_tb_ever	1	58.358	-16103
- random_arm	1	58.359	-16103
- employment_status	2	58.383	-16102
<none>		58.354	-16102
- marital_status	3	58.421	-16099
- gender	1	58.420	-16095

Step: AIC=-16103.8

```
hiv_status_current ~ random_arm + dx_heartdisease_ever + dx_tb_ever +
  employment_status + gender + marital_status + religious_affil
```

	Df	Deviance	AIC
- religious_affil	1	58.357	-16105
- dx_heartdisease_ever	1	58.357	-16105
- dx_tb_ever	1	58.359	-16105
- random_arm	1	58.360	-16105
- employment_status	2	58.383	-16104
<none>		58.354	-16104
- marital_status	3	58.421	-16101
- gender	1	58.429	-16096

Step: AIC=-16105.42

```
hiv_status_current ~ random_arm + dx_heartdisease_ever + dx_tb_ever +
  employment_status + gender + marital_status
```

	Df	Deviance	AIC
- dx_heartdisease_ever	1	58.360	-16107
- dx_tb_ever	1	58.361	-16107
- random_arm	1	58.363	-16107
<none>		58.357	-16105
- employment_status	2	58.387	-16105
- marital_status	3	58.427	-16102
- gender	1	58.429	-16098

Step: AIC=-16107

```
hiv_status_current ~ random_arm + dx_tb_ever + employment_status +
  gender + marital_status
```

	Df	Deviance	AIC
- dx_tb_ever	1	58.365	-16108

```

- random_arm          1    58.366 -16108
<none>                58.360 -16107
- employment_status   2    58.391 -16107
- marital_status      3    58.432 -16103
- gender              1    58.432 -16099

```

Step: AIC=-16108.41

```

hiv_status_current ~ random_arm + employment_status + gender +
  marital_status

```

```

          Df Deviance    AIC
- random_arm      1    58.370 -16110
<none>            58.365 -16108
- employment_status 2    58.396 -16108
- marital_status    3    58.437 -16105
- gender           1    58.437 -16101

```

Step: AIC=-16109.63

```

hiv_status_current ~ employment_status + gender + marital_status

```

```

          Df Deviance    AIC
<none>            58.370 -16110
- employment_status 2    58.403 -16109
- marital_status    3    58.443 -16106
- gender           1    58.442 -16102

```

```

Call: glm(formula = hiv_status_current ~ employment_status + gender +
  marital_status, data = bcpp)

```

Coefficients:

```

              (Intercept)
              0.0190786
employment_statusUnemployed looking for work
              0.0045117
employment_statusUnemployed not looking for work
              0.0006411
gender
              -0.0063009
marital_statusMarried
              -0.0162061
marital_statusSingle/never married
              -0.0098927
marital_statusWidowed
              -0.0201614

```

Degrees of Freedom: 7825 Total (i.e. Null); 7819 Residual

Null Deviance: 58.56
Residual Deviance: 58.37 AIC: -16110

```
# automated backward selection chooses glm(formula = hiv_status_current ~  
  ↪ employment_status + gender + marital_status, data = bcpp), but this excludes  
  ↪ the random_arm variable crucial to answering our research question  
  
# Stepwise selection using AIC as the metric  
step(intercept_glm, scope = list(lower = intercept_glm, upper = full_model),  
  ↪ direction="both")
```

Start: AIC=-16096.11
hiv_status_current ~ random_arm

	Df	Deviance	AIC
+ gender	1	58.480	-16103
+ marital_status	3	58.475	-16100
+ employment_status	2	58.497	-16099
+ education	4	58.481	-16097
+ age_at_interview	1	58.530	-16096
<none>		58.546	-16096
+ dx_tb_ever	1	58.540	-16095
+ dx_heartdisease_ever	1	58.541	-16095
+ smoke	1	58.542	-16095
+ partners_lifetime	1	58.545	-16094
+ religious_affil	1	58.545	-16094
+ alcohol	4	58.522	-16091

Step: AIC=-16102.92
hiv_status_current ~ random_arm + gender

	Df	Deviance	AIC
+ marital_status	3	58.396	-16108
+ employment_status	2	58.437	-16105
+ age_at_interview	1	58.455	-16104
+ education	4	58.413	-16104
<none>		58.480	-16103
+ religious_affil	1	58.470	-16102
+ dx_heartdisease_ever	1	58.474	-16102
+ dx_tb_ever	1	58.475	-16102
+ smoke	1	58.479	-16101
+ partners_lifetime	1	58.480	-16101
+ alcohol	4	58.437	-16101
- gender	1	58.546	-16096

Step: AIC=-16108.25
hiv_status_current ~ random_arm + gender + marital_status

	Df	Deviance	AIC
+ employment_status	2	58.365	-16108
<none>		58.396	-16108
+ religious_affil	1	58.391	-16107
+ dx_tb_ever	1	58.391	-16107
+ dx_heartdisease_ever	1	58.392	-16107
+ education	4	58.348	-16107
+ age_at_interview	1	58.395	-16106
+ smoke	1	58.396	-16106
+ partners_lifetime	1	58.396	-16106
+ alcohol	4	58.359	-16105
- marital_status	3	58.480	-16103
- gender	1	58.475	-16100

Step: AIC=-16108.41

hiv_status_current ~ random_arm + gender + marital_status + employment_status

	Df	Deviance	AIC
<none>		58.365	-16108
- employment_status	2	58.396	-16108
+ dx_tb_ever	1	58.360	-16107
+ dx_heartdisease_ever	1	58.361	-16107
+ religious_affil	1	58.362	-16107
+ age_at_interview	1	58.364	-16106
+ smoke	1	58.365	-16106
+ partners_lifetime	1	58.365	-16106
+ education	4	58.327	-16106
+ alcohol	4	58.332	-16105
- marital_status	3	58.437	-16105
- gender	1	58.437	-16101

Call: glm(formula = hiv_status_current ~ random_arm + gender + marital_status +
employment_status, data = bcpp)

Coefficients:

```

(Intercept)
0.0200285
random_arm
-0.0017262
gender
-0.0063023
marital_statusMarried
-0.0162065
marital_statusSingle/never married
-0.0099330

```

```

marital_statusWidowed
-0.0201497
employment_statusUnemployed looking for work
0.0043805
employment_statusUnemployed not looking for work
0.0006099

```

```

Degrees of Freedom: 7825 Total (i.e. Null); 7818 Residual
Null Deviance: 58.56
Residual Deviance: 58.36 AIC: -16110

```

```

# stepwise selection using AIC chooses glm(formula = hiv_status_current ~
  ↪ random_arm + gender + marital_status + employment_status, data = bcpp)

# To use BIC as the metric instead, we can set k = log(n)
n <- nrow(bcpp)
step(intercept_glm, scope = list(lower = intercept_glm, upper = full_model),
  ↪ direction="both", k = log(n))

```

```

Start: AIC=-16082.18
hiv_status_current ~ random_arm

```

	Df	Deviance	AIC
<none>		58.546	-16082
+ gender	1	58.480	-16082
+ age_at_interview	1	58.530	-16075
+ dx_tb_ever	1	58.540	-16074
+ dx_heartdisease_ever	1	58.541	-16074
+ smoke	1	58.542	-16074
+ partners_lifetime	1	58.545	-16073
+ religious_affil	1	58.545	-16073
+ employment_status	2	58.497	-16071
+ marital_status	3	58.475	-16065
+ education	4	58.481	-16055
+ alcohol	4	58.522	-16050

```

Call: glm(formula = hiv_status_current ~ random_arm, data = bcpp)

```

```

Coefficients:
(Intercept) random_arm
0.008605 -0.002154

```

```

Degrees of Freedom: 7825 Total (i.e. Null); 7824 Residual
Null Deviance: 58.56
Residual Deviance: 58.55 AIC: -16100

```

```

# chooses Call: glm(formula = hiv_status_current ~ random_arm, data = bcpp)

# First step of the backward selection process
# full_model <- glm(hiv_status_current ~ random_arm + age_at_interview + alcohol
  ↪ + dx_heartdisease_ever + dx_tb_ever + education + employment_status + gender
  ↪ + marital_status + partners_lifetime + religious_affil + smoke, test = "F")
# drop1(full_model, test = "F")
# this test drops the random_arm variable that is crucial to include in order to
  ↪ answer our research question

# Evaluating AIC of different models with ad without interactions
mod_1 <- glm(hiv_status_current ~ random_arm, data = bcpp)
mod_2 <- glm(hiv_status_current ~ random_arm + gender, data = bcpp)
mod_2.5 <- glm(hiv_status_current ~ random_arm * gender, data = bcpp)
mod_3 <- glm(hiv_status_current ~ random_arm + gender + marital_status, data =
  ↪ bcpp)
mod_3.5 <- glm(hiv_status_current ~ random_arm + gender * marital_status, data =
  ↪ bcpp)
mod_4 <- glm(hiv_status_current ~ random_arm + gender + marital_status +
  ↪ employment_status, data = bcpp)
mod_4.5 <- glm(hiv_status_current ~ random_arm + gender + marital_status *
  ↪ employment_status, data = bcpp)
mod_4.55 <- glm(hiv_status_current ~ random_arm + gender * marital_status +
  ↪ employment_status, data = bcpp)
mod_4.6 <- glm(hiv_status_current ~ random_arm + gender * employment_status +
  ↪ marital_status, data = bcpp)
mod_4.65 <- glm(hiv_status_current ~ random_arm + gender * marital_status *
  ↪ employment_status, data = bcpp)
mod_4.7 <- glm(hiv_status_current ~ random_arm * gender + marital_status +
  ↪ employment_status, data = bcpp)
mod_4.75 <- glm(hiv_status_current ~ random_arm * gender * marital_status +
  ↪ employment_status, data = bcpp)
mod_4.755 <- glm(hiv_status_current ~ random_arm * marital_status + gender *
  ↪ employment_status, data = bcpp)
mod_4.8 <- glm(hiv_status_current ~ random_arm * marital_status + gender +
  ↪ employment_status, data = bcpp)
mod_4.85 <- glm(hiv_status_current ~ random_arm * employment_status + gender +
  ↪ marital_status, data = bcpp)

# AIC BIC
mod_1 |> glance()

```

```
# A tibble: 1 x 8
```

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	58.6	7825	8051.	-16096.	-16075.	58.5	7824	7826

```
# -16096.11,      -16075.22
```

```
mod_2 |> glance()
```

```
# A tibble: 1 x 8
```

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	58.6	7825	8055.	-16103.	-16075.	58.5	7823	7826

```
# -16102.92 ,      -16075.06
```

```
# with gender interaction
```

```
mod_2.5 |> glance()
```

```
# A tibble: 1 x 8
```

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	58.6	7825	8056.	-16102.	-16068.	58.5	7822	7826

```
# -16102.45 ,      -16067.62
```

```
mod_3 |> glance()
```

```
# A tibble: 1 x 8
```

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	58.6	7825	8061.	-16108.	-16059.	58.4	7820	7826

```
# -16108.25,      -16059.5
```

```
# with gender marital status interaction
```

```
mod_3.5 |> glance()
```

```
# A tibble: 1 x 8
```

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	58.6	7825	8062.	-16104.	-16034.	58.4	7817	7826

```
# -16103.97,      -16034.32
```

```
#-----
```

```
# lowest AIC is with this model:
```

```
# glm(hiv_status_current ~ random_arm + gender + marital_status +
  ↪ employment_status, data = bcpp)
mod_4 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int> <dbl>    <dbl>    <dbl>    <dbl>      <int> <int>
1         58.6     7825  8063. -16108. -16046.     58.4        7818  7826
```

```
# -16108.41 ,      -16045.72
#-----
# marital status and employment status interaction
mod_4.5 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int> <dbl>    <dbl>    <dbl>    <dbl>      <int> <int>
1         58.6     7825  8066. -16103. -15998.     58.3        7812  7826
```

```
# -16102.53,      -15998.05
#marital status and gender interaction
mod_4.55 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int> <dbl>    <dbl>    <dbl>    <dbl>      <int> <int>
1         58.6     7825  8064. -16104. -16021.     58.4        7815  7826
```

```
# -16104.2,      -16020.62
# gender and employment status interaction
mod_4.6 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int> <dbl>    <dbl>    <dbl>    <dbl>      <int> <int>
1         58.6     7825  8063. -16105. -16028.     58.4        7816  7826
```

```
# -16104.9,      -16028.28
# interactions between gender and marital status and employment status
mod_4.65 |> glance()
```



```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8069. -16085. -15904.    58.3       7801  7826
```

```
# -16085.22,      -15904.13
```

```
#interaction between treatment and gender
mod_4.7 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8064. -16108. -16038.    58.4       7817  7826
```

```
# -16107.98,      -16038.33
```

```
# interaction between treatment, gender, and marital status
mod_4.75 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8069. -16100. -15968.    58.3       7808  7826
```

```
# -16100.41,      -15968.07
```

```
# interaction between treatment and marital status, and between gender and
  ↪ employment status
mod_4.755 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8066. -16104. -16007.    58.3       7813  7826
```

```
# -16104.49,      -16006.98
```

```
#interaction between treatment and marital status
mod_4.8 |> glance()
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8066. -16108. -16024.    58.3       7815  7826
```

```
# -16108.03,          -16024.45

#interaction between treatment and employment status
mod_4.85 |> glance()

# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual  nobs
      <dbl>    <int>  <dbl>   <dbl>   <dbl>   <dbl>      <int> <int>
1      58.6     7825  8063. -16105. -16028.    58.4       7816  7826

# -16104.98,          -16028.37

# treatment community group summary
random_arm_stats <- table(bcpp$random_arm)
random_arm_df <- data.frame(t(as.vector(random_arm_stats)))
colnames(random_arm_df) <- names(random_arm_stats)
kable(random_arm_df, caption = "Summary Statistics for Study Participant
  ↪ Intervention Community", format = "html")
```

Table 9: Summary Statistics for Study Participant Intervention Community

	0	1
	3951	3875

```
#gender summary
gender_stats <- table(bcpp$gender)
gender_df <- data.frame(t(as.vector(gender_stats)))
colnames(gender_df) <- names(gender_stats)
kable(gender_df, caption = "Summary Statistics for Study Participant Gender",
  ↪ format = "html")
```

Table 10: Summary Statistics for Study Participant Gender

	0	1
	4764	3062

```
# marital status summary
marital_status_stats <- table(bcpp$marital_status)
marital_status_df <- data.frame(t(as.vector(marital_status_stats)))
colnames(marital_status_df) <- names(marital_status_stats)
kable(marital_status_df, caption = "Summary Statistics for Study Participant
  ↪ Marital Status", format = "html")
```

Table 11: Summary Statistics for Study Participant Marital Status

Divorced/separated	Married	Single/never married	Widowed
107	1346	6162	211

```
# employment status summary
employment_status_stats <- table(bcpp$employment_status)
employment_status_df <- data.frame(t(as.vector(employment_status_stats)))
colnames(employment_status_df) <- names(employment_status_stats)
kable(employment_status_df, caption = "Summary Statistics for Study Participant
↪ Employment Status", format = "html")
```

Table 12: Summary Statistics for Study Participant Employment Status

Employed	Unemployed looking for work	Unemployed not looking for work
2317	3224	2285

```
# age at interview summary stats
age_at_interview_stats <- summary(bcpp$age_at_interview)
age_at_interview_df <- data.frame(t(as.vector(age_at_interview_stats)))
colnames(age_at_interview_df) <- names(age_at_interview_stats)
kable(age_at_interview_df, caption = "Summary Statistics for Study Participant
↪ Age at Interview", format = "html")
```

Table 13: Summary Statistics for Study Participant Age at Interview

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
16	22.4	29.7	33.55229	43.1	64.9

```
# HIV status summary
hiv_status_current_stats <- table(bcpp$hiv_status_current)
hiv_status_current_df <- data.frame(t(as.vector(hiv_status_current_stats)))
colnames(hiv_status_current_df) <- names(hiv_status_current_stats)
kable(hiv_status_current_df, caption = "Summary Statistics for Study Participant
↪ HIV Status One Year in Trial", format = "html")
```

Table 14: Summary Statistics for Study Participant HIV Status One Year in Trial

0	1
7767	59

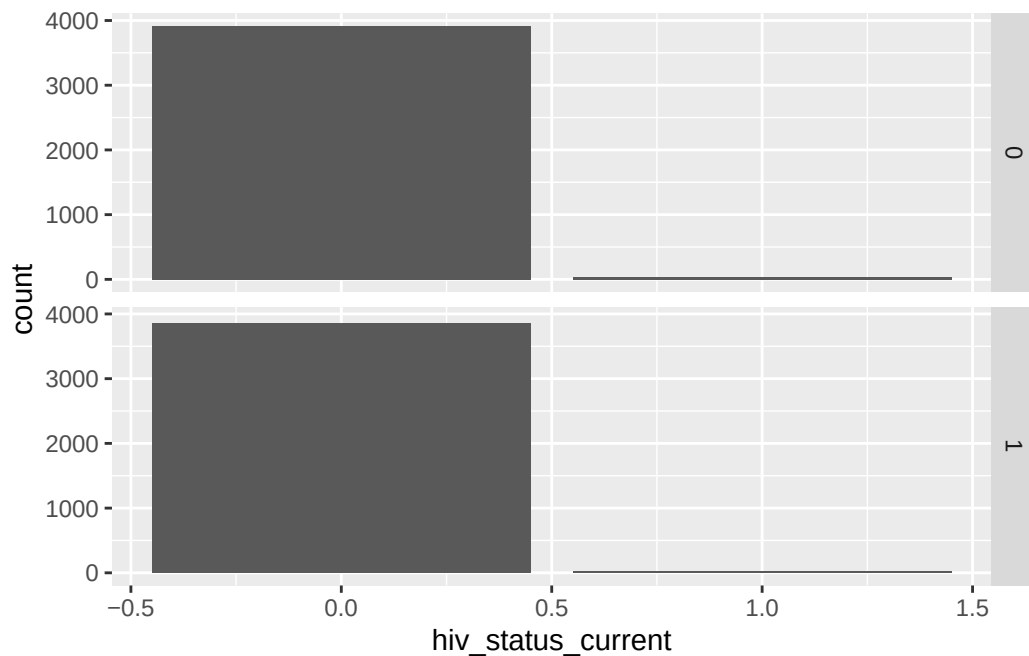
```
# creating a table for how many people in the treatment community vs the standard
  ↪ of care community contracted HIV
table1 <- bcpp |>
  group_by(random_arm) |>
  filter(hiv_status_current == "HIV-uninfected") |>
  summarise("# uninfected" = n())

table2 <- bcpp |>
  group_by(random_arm) |>
  filter(hiv_status_current == "HIV-infected") |>
  summarise("# infected" = n())

table3 <- full_join(table1, table2, by = "random_arm")
kable(table3)
```

random_arm	# uninfected	# infected
------------	--------------	------------

```
#visualization for the treatment/HIV status distribution by gender
ggplot(bcpp, aes(x = hiv_status_current, fill = gender)) +
  geom_histogram(stat="count") +
  facet_grid(random_arm ~ .)
```



```
table4 <- bcpp |>
  group_by(hiv_status_current) |>
```

```

filter(gender == "M") |>
summarise("# M" = n())

table5 <- bcpp |>
  group_by(hiv_status_current) |>
  filter(gender == "F") |>
  summarise("# F" = n())

table6 <- full_join(table4, table5, by = "hiv_status_current")
kable(table6)

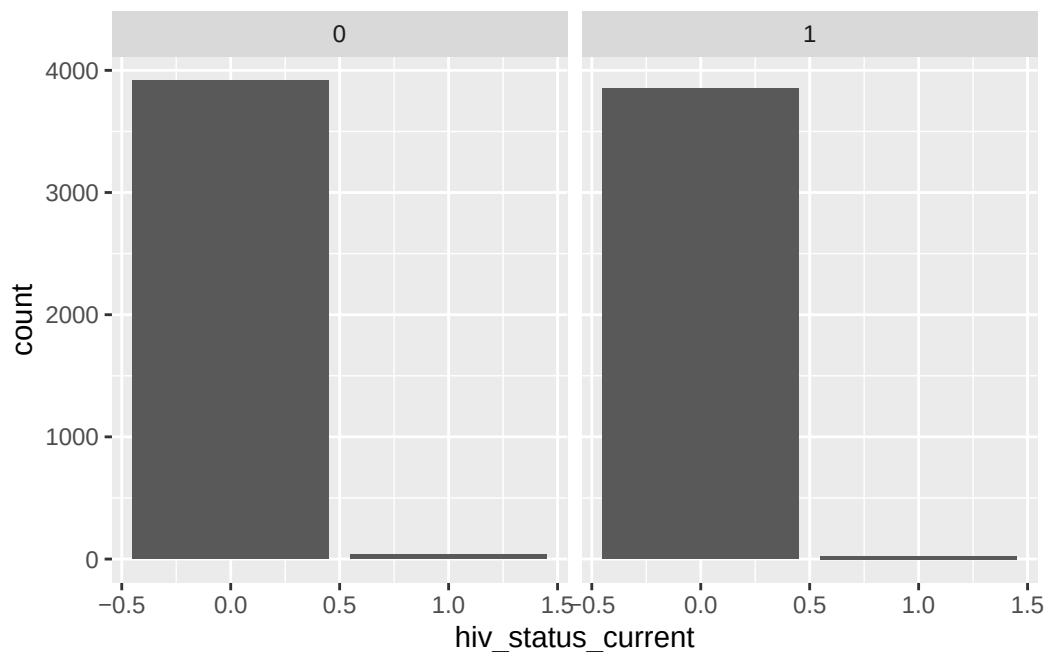
```

hiv_status_current	# M	# F
--------------------	-----	-----

```

#visualization for the treatment/HIV status distribution by gender
ggplot(bcpp, aes(x = hiv_status_current, fill = gender)) +
  geom_histogram(stat="count") +
  facet_wrap(random_arm ~ .)

```



```

#Answering Question 1: How do the odds of contracting HIV change depending on
  ↳ whether or not study participants live in an intervention community?

# Fitting the logistic regression line for hiv status as a function of treatment
  ↳ or standard community
og_model <- glm(hiv_status_current ~ random_arm, data = bcpp)

```

```
# Representing the logistic regression table as a dataframe (i.e., tidying the
  ↳ summary() output)
og_model_table <- summary(og_model)$coefficients

# Creating a nicely formatted table from the dataframe using the kable package
og_model_table |> kbl(col.names = c("Estimate", "Std. Error", "t value",
  ↳ "Pr(>|t|)"), align = "c", booktabs = T, linesep="", digits = c(6, 6, 4, 4))
  ↳ |> kable_classic(full_width = F, latex_options = c("HOLD_position"))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.008605	0.001376	6.2530	0.0000
random_arm	-0.002154	0.001956	-1.1013	0.2708

```
# Calculating the odds of contracting HIV depending on community
exp(-0.002154)
```

```
[1] 0.9978483
```

```
(1-0.9978483)*100
```

```
[1] 0.21517
```

```
#Answering Question 2: What does this relationship look like after adjusting for
  ↳ other important predictors of HIV infection? And answering question 3 with
  ↳ the p-value for random_arm coefficient estimate

# Fitting the logistic regression line for hiv status as a function of treatment
  ↳ or standard community, with the model found with lowest AIC through step-wise
  ↳ selection
new_model <- glm(hiv_status_current ~ random_arm + gender + marital_status +
  ↳ employment_status, data = bcpp)

# Representing the logistic regression table as a dataframe (i.e., tidying the
  ↳ summary() output)
new_model_table <- summary(new_model)$coefficients

# Creating a nicely formatted table from the dataframe using the kable package
new_model_table |> kbl(col.names = c("Estimate", "Std. Error", "t value",
  ↳ "Pr(>|t|)"), align = "c", booktabs = T, linesep="", digits = c(6, 6, 4, 4))
  ↳ |> kable_classic(full_width = F, latex_options = c("HOLD_position"))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.020029	0.008556	2.3409	0.0193
random_arm	-0.001726	0.001959	-0.8813	0.3782
gender	-0.006302	0.002028	-3.1070	0.0019
marital_statusMarried	-0.016207	0.008679	-1.8673	0.0619
marital_statusSingle/never married	-0.009933	0.008438	-1.1772	0.2392
marital_statusWidowed	-0.020150	0.010270	-1.9620	0.0498
employment_statusUnemployed looking for work	0.004380	0.002391	1.8319	0.0670
employment_statusUnemployed not looking for work	0.000610	0.002565	0.2378	0.8121

```
# Calculating the odds of contracting HIV depending on community
exp(-0.0017262)
```

```
[1] 0.9982753
```

```
(1-0.9982753)*100
```

```
[1] 0.17247
```